

## Article

# Machine Learning as a Tool for Assessment and Management of Fraud Risk in Banking Transactions

Antonio Dichev , Silvia Zarkova \*  and Petko Angelov 

Department of Finance and Credit, Tsenov Academy of Economics, 5250 Svishtov, Bulgaria;  
antoniodichev@yahoo.com (A.D.); p.angelov@uni-svishtov.bg (P.A.)

\* Correspondence: s.zarkova@uni-svishtov.bg

**Abstract:** The present work aims to fill the gaps in existing research on the application of machine learning in fraud detection and management in the banking sector. It provides a theoretical perspective on the evolution of algorithms, highlights practical aspects, and derives relevant metrics for evaluating their performance on unbalanced data. In the growing context of artificial intelligence, the adoption of an innovative, systematic approach to studying fraud in banking transactions through advanced machine learning algorithms is completely positive for the overall accuracy and effectiveness of risk management and has really practical and applied significance. The proven methodology (Classification and Regression Trees, Gradient Boosting, and Extreme Gradient Boosting) was tested on nearly 1.5 million in the banking sector, confirming the observations related to the application of fundamental assessments and specialized statistical methods through machine learning algorithms, demonstrating superior discriminatory power compared to classical models. The development provides valuable insights for managers, researchers, and policymakers aiming to strengthen the security and resilience of banking systems in times of evolving financial fraud challenges.

**Keywords:** machine learning; risk management; bank transaction; fraud detection



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## 1. Introduction

In recent years, the banking industry has been reminded once again that ensuring customer trust is the most critical element for its survival. After the bankruptcy of Silicon Valley Bank (SVB), which triggered the second-largest shock to the U.S. banking system since the 2008 crisis, the importance of banks ensuring customer safety and trust was reaffirmed. Banking crises have a destabilizing impact on financial markets and can hamper global economic growth, but one of the main challenges faced by banking institutions in the era of ever-evolving technology and artificial intelligence is the increasing risk of fraud. Financial institutions rely on risk assessment to manage risk and prevent errors. The materialization of financial risks, including fraud and mistakes, often leads to missed opportunities, losses, financial turmoil, and a loss of trust.

When it comes to risk management, accurate assessment through an automated approach is essential. The scope of risk management also includes fraud in bank transactions. Artificial intelligence (AI) technologies offer modern systems powerful tools for identifying, assessing, and managing risk. The implementation of these technologies in the banking sector often highlights significant added value, with an emphasis on a higher level of efficiency and accuracy in assessments.

This paper presents an analysis of the added value in identifying fraudulent bank transactions through the application of some modern and sophisticated machine learning

algorithms: Classification and Regression Trees (CART), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost). The results were compared with a representative of the so-called traditional methods and models commonly used for similar purposes: Logistic Regression. The sample used for the study consists of 1,498,177 observations from a synthetic database provided by J. P. Morgan<sup>1</sup>, which remains their exclusive property.

Based on this foundation, the object of the study is the assessment of the fraud risk in bank transactions. The subject of the study focuses on the methods for quantitative modeling and the construction of tools for identifying fraud in bank transactions as an element of fraud prevention and the broader risk management framework. The main purpose is to examine the added value of machine learning in the automated identification of fraud in bank transactions. The research hypothesis asserts that the application of an innovative, systematic approach to the study of fraud in bank transactions through the development of a quantitative methodology based on advanced machine learning algorithms contributes positively to the overall accuracy and effectiveness of risk management frameworks.

The present work aims to fill the gaps in existing research on the application of machine learning in fraud detection and management in the banking sector. The advantages of specialized assessment methods based on artificial intelligence for high-frequency data in financial institutions, such as banking transactions, are highlighted. The development provides valuable insights for managers, researchers, and policymakers aiming to strengthen the security and resilience of banking systems in times of evolving financial fraud challenges.

## 2. Theoretical Framework on the Topic

### 2.1. General Understanding of Machine Learning (ML) as a Risk Management Tool

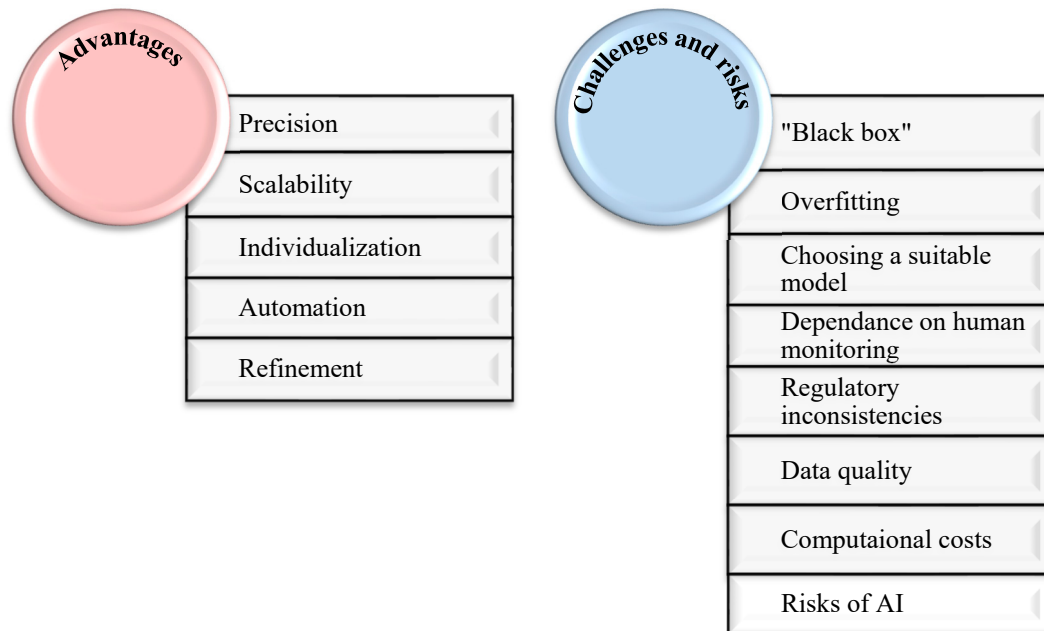
Artificial intelligence offers a wide range of technological solutions, including machine learning, natural language processing, expert systems, robotics, and more. Machine learning theory is thematically related to the fundamental theories and practical applications of learning from data as a computational process. It seeks to understand at a high level what mathematical abilities and information are needed to successfully learn different types of tasks and to understand the basic algorithmic principles involved in making computers improve performance with feedback (Blum, 2025). The goals of this theory are both to help design better methods for automated learning and to understand the fundamental problems in this learning process. Within the framework of testing and communization, the theory of machine learning, statistical solutions, optimization, and algorithmic decision making are combined.

Machine learning, as a subset of artificial intelligence technologies, has been revolutionizing the financial sector by offering opportunities for customized banking solutions using big data. The potential of this toolbox is also effectively applied in risk management. The following definition of machine learning (ML) is commonly used in many accessible resources:

“A computer program is said to learn from experience  $E$  with respect to some class of tasks  $T$  and performance measure  $P$ , if its performance of tasks in  $T$ , as measured by  $P$ , improves with expanded experience  $E$ .” (Mitchell, 1997, p. 2)

Although logistic regression meets the definition above and theoretically falls within the scope of machine learning, this study treats logistic regression as an alternative to machine learning algorithms. It is considered a representative of traditional algorithms and methods for modeling the probability of fraud. This distinction is made because in practice, logistic regression is often not perceived and categorized as an ML algorithm. Furthermore, this division aligns with ongoing regulatory and institutional discussions within the banking sector (EBA, 2021).

Machine learning-based risk assessment and management solutions offer several key advantages (Figure 1): enhanced accuracy and precision in deriving non-linear effects between scenarios and factors; the ability to process large-scale datasets, uncover hidden relationships, and detect interactions that are not immediately apparent; personalization of system models based on individual clients' investment, credit, and financial behavior; automated detection of financial anomalies; improved productivity in dynamic environments with the capacity of continuous improvement.



**Figure 1.** Advantages vs. challenges and risks of machine learning. Source: Developed by the authors.

Along with their advantages in practice, machine learning (ML) algorithms present certain challenges and risks, the most significant of which include overfitting; the “black box” issue; the application of algorithms requires the use of large datasets, which raises concerns about data confidentiality and quality; the wide range of available ML algorithms, making the selection of an appropriate model a challenge; the need for human supervision by individuals with appropriate skills to ensure the ethical and strategic governance of financial institutions; compliance with regulatory requirements that demand specific controls, data transparency, and the ability to perform ex-post audits.

The two main concerns that are discussed in greater detail in this research are as follows:

- **Overfitting**—machine learning algorithms are easy to overfit because they learn patterns from training data, and if the model is too complex (e.g., too many parameters or deep trees), it can memorize noise rather than generalizable trends. Overfitting leads to a high accuracy on training data but poor performance on testing data which is reducing the model’s ability to generalize. To prevent overfitting, the current research primarily relies on maintaining a separate sample of unseen data (validation/testing sample) and implementing a structured training approach. Further details are provided in the following sections. The “Black Box” problem—refers to the lack of interpretability and transparency in how complex models, especially deep learning and ensemble methods, make decisions. Some models, such as deep neural networks or gradient-boosted trees, can achieve high accuracy but operate in ways that are difficult for humans to understand. Although it is not the focus of the current research, we acknowledge the importance of this aspect and consider it worthy of further analysis. Approaches with a strong track record in both practice and the literature, such as

feature importance and SHAP values (Lundberg & Lee, 2017), are potential methods for application and interpretation.

## 2.2. Fraud Risk Management in the Banking Sector

“The financial sector has a fundamental role in the modern economy of the 21st century, ensuring the transfer of funds. One of the brightest constituent units of the financial sector are banks, which, through their activities related to “public attraction of deposits or other repayable funds and provide loans or other financing at their own expense and at their own risk” (BNB, 2023). Traditionally, statistical approaches and mathematical methods have been employed to analyze and extract information from systematized datasets to derive strategic insights, assess investments, evaluate creditworthiness, and mitigate fraud risks, among others. Technological advances have significantly enhanced the ability of modern banking institutions to detect, prevent, and mitigate fraud, as well as to manage broader risks effectively.

Fraud in the banking sector remains a critical aspect of risk management, as it can impact financial performance, institutional reputation, and customer trust. According to the classification of (Basel Committee on Banking Supervision, 2023), fraud can be categorized into four groups: unauthorized payment transactions, manipulation of payers to issue fraudulent payment orders, fraud related to other banking products used by customers, fraud targeting customer data or banking systems. Common types of bank fraud include forgery, check fraud, credit card fraud, identity theft, phishing, and more (Fraud, 2024).

As a form of financial crime, fraud involves exploiting financial institutions or their services for personal gain or facilitating other criminal activities. Thus, timely identification and prevention of fraud are essential to minimize its impact. This can be achieved through traditional statistical methods, as well as through the integration of ML methods aimed at increasing the efficiency of processes, reducing human errors, and increasing adaptability to changing economic conditions (Pavlova-Banova & Stoyanova, 2024).

The above highlights the significant challenges faced by the banking sector due to potential fraud and abuse. To address these challenges and assess risks effectively in today’s dynamic environment, modern solutions for fraud risk assessment and management, particularly those leveraging artificial intelligence and machine learning algorithms, can be employed. These advanced technologies offer banks a powerful opportunity to enhance their ability to detect, prevent, and mitigate fraud, ensuring greater security and operational resilience.

## 3. Review of Literature Sources

In modern research, a range of studies explores the application of open-source software and machine learning for risk assessment within the financial sector, particularly in fraud identification and prevention in the banking sector. In this regard, a comprehensive review of the literature has been conducted, highlighting key aspects of the application of machine learning in the financial sector. The aim of the analysis is to explore good practices and identify gaps in the literature, which we will try to compensate for. One of the latest research works in the field is the development of (Khaled Alarfaj & Shahzadi, 2025). They apply deep learning to fraud detection using graph neural networks and autoencoders to improve business practices and reduce fraudulent activities in large organizations. The findings show that these methods effectively detect fraud with a balance of precision and recall, improving the efficiency of banking systems. This study concludes that by using deep learning algorithms, online credit card fraud detection in banks can be controlled, and the efficiency of the banking system can be improved. Fraud activities can also be managed in real time and on dynamic data sets using deep learning algorithms, which allows for

continuous improvement of the fraud detection and prevention system. [Khang \(2025\)](#) explores the importance of artificial intelligence (AI)-based predictive analytics tools in the financial services industry and their role in combating financial fraud. The author shares the problem of improving fraud detection and prevention through the application of predictive analytics. Through his work, he contributes to a deeper understanding of the importance of predictive analytics in the field of finance and its fundamental role in cybersecurity and the fight against fraud. Analytics is transforming not only the field of finance but also banking, as it increases the transparency of lending activities. [Kumar et al. \(2024\)](#) focus on the explosive growth in the use of credit cards for online purchases, which in turn has led to an increase in fraud. The authors present machine learning methods as essential for identifying fraudulent transactions. The paper describes an intelligent approach for detecting and predicting online fraud transactions using an extreme gradient boosting algorithm (XGBoost). The model predicts whether a given transaction is fraudulent or not. [Alagic et al. \(2024\)](#), present this paradigm and examine the potential to predict credit risk in the banking sector based on a survey of nearly two thousand people. The results of the tested algorithms demonstrate that XGBoost achieved the highest accuracy at 84%, surpassing Gradient Boosting (83%) and K-nearest neighbors (KNN) algorithms (83%). On the second dataset, the Decision Tree algorithm reached 85% accuracy, followed by Decision Tree and K-nearest neighbors (KNN) algorithms (83%). At the same time, [Raúl et al. \(2024\)](#) conducted a bibliometric analysis to explore the progress, challenges, and applications of artificial intelligence (AI) and machine learning (ML) in the banking sector through the consideration of variables related to the distribution of publications and the analysis of keyword dynamics. [Kumar et al. \(2024\)](#) focus on the explosive growth in the use of credit cards for online purchases, which in turn has led to an increase in fraud. The authors present machine learning methods as essential for identifying fraudulent transactions. The paper describes an intelligent approach for detecting and predicting online fraud transactions using an extreme gradient boosting algorithm (XGBoost). The model predicts whether a given transaction is fraudulent or not. [Zhelev and Kostova \(2024\)](#) examines the extent to which the implementation of digital tools in financial control and the risk of fraud improves its effectiveness. The study highlights the need for continued investment in digital technologies and training to maximize the benefits of their application in financial control. The importance of the application of open-source software is also explored by the authors ([Zaharieva et al., 2022](#)), implementing it in capital market analysis. For its part, [Majumder et al. \(2024\)](#) use three machine learning (ML) evaluation models in a comparative aspect, namely Extra Trees Classifier (ETC), Support Vector Machine Classifier (SVMC), and Multinomial Naive Bayes Classifier (MNBC), and performs a technical comparative analysis with respect to seven performance indicators in the transport sector. [Achary and Shelke \(2023\)](#) propose an AI-based approach to enhance the detection of fraud in banking transactions. They emphasize the critical role of adequate data and address data imbalances by employing the oversampling technique for synthetic minority classes. Using the XGBoost algorithm, they demonstrate a model performance exceeding 95%. [Arunkumar et al. \(2023\)](#) use machine learning (Random Forest Algorithm and Support Vector Algorithm) to predict loan approval and detect credit card fraud. In this case, 92% accuracy was achieved using the Random Forest algorithm. [Raj and Vetrithangam \(2023\)](#) apply ML techniques to predict customer churn as a major risk for revenue loss and one of the key issues for large institutions. [Iu \(2023\)](#) highlights the growing integration of banking activities in a digital environment, which leads to a substantial rise in bank fraud. In his work, the author confirms the importance of applying artificial intelligence in the study of bank fraud. For ([Rode et al., 2022](#)), fraud in bank transactions is a very common problem, the prevention of which requires the application of modern machine learning technologies.



Algorithms can reduce the need for manual reviews and analysis by employees. The theoretical review of the authors' team explores the practical aspects and opportunities for the future use of ML for fraud detection purposes (Guerra et al., 2022) and investigates the ability to model risk by comparing a series of well-established machine learning algorithms with a classical statistical model. By applying the XGBoost algorithm to real data for the period from 2014 to March 2021, they demonstrate the model's superiority of the model and potential for upgrades, such as stress testing and early warning systems. Zahariev et al. (2022) focus on the identification and metrification of the macroeconomic drivers of bank profitability of Bulgarian banks through the application of an iterative testing classical methodology through VECM and SVR. As final results, they conclude that with the growing regulatory requirements for the formation of buffers against macroprudential risks, banks report a relatively neutral dependence of ROA and ROE (Prodanov et al., 2022) on the influence of macroeconomic factors, which is logically compensated by effective management decisions and innovations in interbank competition. Madaan et al. (2020), through the evaluation of certain indicators, present two machine learning models aimed at predicting the risk of insolvency of borrowers. The algorithms they train on the same dataset in a comparative aspect are Random Forest and Decision Trees. As a result, it is observed that Random Forest outperforms the Decision Trees algorithm by nearly 10%. Finally, Carmona et al. (2019) employ XGBoost to predict insolvency within the U.S. banking sector, analyzing data from 156 national commercial banks. On this basis, it is found that lower values of retained earnings-to-average equity ratios, return on assets, and risk-based total capital ratios are indicators of increased bank failure risk.

The theoretical review confirms the relevance of the work presented in this paper. While there are several studies on the use of machine learning (ML) for fraud detection and management, many of them lack a theoretical framework that helps track the evolution of algorithms or fail to highlight the key differences between them. Additionally, practical aspects are often overlooked, and in some cases, metrics like accuracy, which are not suitable for imbalanced datasets, are used to evaluate performance. This paper aims to address these gaps by providing a theoretical perspective on algorithm evolution, emphasizing practical aspects, and using more appropriate evaluation metrics for imbalanced data. From a theoretical and practical aspect, it is demonstrated that machine learning methods are a vital and contemporary component of modern banking operations, particularly in risk assessment and the detection and prevention of bank fraud. By applying and evaluating these methods, financial and reputational risks can be mitigated, and customer trust can be preserved. The integration of such innovative approaches and their timely adaptation to evolving conditions could be a key part of an effective risk mitigation framework within the financial sector, especially in banking.

#### 4. Description of the Dataset and Methodology

This study utilizes a synthetic database provided by J. P. Morgan<sup>2</sup>. The sample contains transaction data and is oriented towards identifying fraudulent activities among them. The dataset contains a diverse range of transaction types, representing both normal activities and unusual or fraudulent activities. The total number of observations is 1,498,177, providing a robust basis for the experimental application of modern and sophisticated machine learning algorithms capable of handling large data sets. The sample is very clean and well-structured, without any issues in that regard.

Synthetic data are a reliable alternative to real data for research and model training. These kinds of data frames are defined as data obtained from a generative process that learns the properties of real data (S. Assefa, 2020). Synthetic data generation is essential in financial services to enable data sharing and analysis while protecting privacy. This

practice allows the creation of realistic datasets that preserve statistical properties while avoiding the exposure of sensitive information.

The dataset contains detailed information, such as data about the sender, the beneficiary, the account, the sector of the sender/beneficiary, the country of the sender/beneficiary, the amount of transaction, the type of transaction, and others. From these raw data, the authors generated the following additional indicators:

- Number of transactions per time frame (week, month);
- Minimum/maximum/total amount of transactions per time interval (week, month);
- Transaction amount/maximum transaction amount per time interval (week, month);
- Transaction amount/average transaction amount per time interval (week, month);
- Transaction type percentage per time interval (week, month), count/amount criteria;
- Client and beneficiary country.

This set of additional indicators is calculated at different levels (account/client) for both the client and the beneficiary. In this way, the authors aimed to capture behavior from multiple perspectives, which can be valuable for identifying unusual patterns and fraudulent transactions. The constructed features are selected to maximize independent insights. For example, while transaction volume captures frequency, ratio-based indicators provide context on relative transaction sizes, helping to distinguish normal variations from anomalies. Furthermore, the analysis of the beneficiary's behavior focuses on their persistence in transactions with other clients and accounts, providing additional insights. This type of construction of additional variables is known in the literature and practice as "feature engineering". This process has a strong importance and significance in modeling most of the challenges in the financial sector. The constructed variables are at different levels (account/client/beneficiary) and across different periods (week/month). Some of the additionally defined indicators include number of transactions, average transaction amount, maximum amount over the transactions, % of transactions per specific type, transaction amount versus average transaction amount, transaction amount versus maximum transaction amount, etc.

The data are further divided into development (75%) and test (25%) samples. The development sample is used during the algorithm training process, while the test sample is utilized to evaluate and validate the model's discriminatory power on data not involved in the training process. The observation selection during the data frame splitting is random, preserving the ratio between the classes of the dependent variable. The preserved unseen data (validation/testing sample) is used as a main instrument against overfitting risk, and all conclusions are based on the results of these data.

#### *4.1. Algorithms and Methods Used to Model the Risk of Fraud in Bank Transactions*

In practice, the risk of a fraudulent transaction is a binary classification problem. Specifically, there are two groups of transactions available, and the purpose of the model is to distinguish between them with the highest possible accuracy. The two defined groups are as follows:

- Class 0—transactions that are typical and do not qualify as fraud;
- Class 1—transactions that are atypical and qualify as fraud.

As a widely used and familiar tool for traditional risk modeling, including detecting fraud in bank transactions, the authors employ logistic regression (LR). Its applied alternatives and representatives of more modern methods and algorithms of machine learning are Classification and Regression Trees (CART), Gradient Boosting (GB), Extreme Gradient Boosting (XGBoost). These machine learning algorithms were chosen based on the following criteria:

- Their extensive use in binary classification tasks, as evidenced by findings in the literature review;
- Their inherent ability to capture non-linear patterns and complex feature interactions;
- Faster training and processing times compared to alternatives like Support Vector Machines and neural networks;
- Strong performance on large datasets with imbalanced classes, offering a balance between interpretability and predictive accuracy;
- They follow an evolutionary in decision tree-based models, with CART as the foundation, Gradient Boosting improving upon it, and XGBoost further optimized the performance through regularization and efficiency enhancements, including the incorporation of second-order derivatives.

The object and scope of this study do not imply a detailed review of the assumptions or specifics of the applied algorithms (Kostova & Zhelev, 2024), but a high-level review is included. It is important to note that both groups of models (traditional models and machine learning algorithms) do not represent an exhaustive list of the appropriate and currently applied methods for predictive classification problems.

As a traditional model, logistic regression is applied in this study. Logistic regression is well-established in theory and widely used in practice. It employs logistic functions to predict the probability of binary outcomes, providing a binary or dichotomous result limited to two possible outcomes. The method analyzes the relationship between one or more independent variables and a dependent variable, making it effective for solving binary classification problems. The main advantage of logistic regression analysis in a machine learning environment is its relatively easy implementation, which does not require significant computation resources. In this study, a logistic regression of the binomial type is applied; that is, the dependent variable has two classes. The linear expression of the regression algorithm is represented as follows (Park, 2013):

$$\text{logit}(y) = \ln\left(\frac{p}{1-p}\right) = \alpha + \beta_1 X_1 + \beta_k X_k \quad (1)$$

The construction of the logistic regression model (1) was carried out using the grouping approach for the independent variables and the so-called “binning” WoE (Weight of evidence) transformation, including the calculation of the IV (information value) indicator (Siddiqi, 2006, pp. 77–81).

In machine learning, there are so-called hyperparameters. These are settings of the algorithm that are not “learned” from the data themselves and that determine its behavior and the complexity of the architecture of the model. That is, even when using the same algorithm (such as XGBoost), it is quite normal for different data samples to result in models with different architecture and complexity. The final values of the hyperparameters are found through a process of searching for the most optimal ones. This is an essential and key step in the overall process of applying ML algorithms. In the most general sense, the rule is to choose the combination of hyperparameter values that leads to the best final outputs. There are different approaches for finding optimal values, such as “grid search”, “random search”, and others. In this study, the so-called “random search” approach was used (Dichev, 2023). By applying this approach, the authors incorporated specific parameters that further contribute to reducing overfitting and enhancing generalization:

- The three-fold cross-validation—ensures that the selected hyperparameters generalize well across different subsets of the training data;
- Histogram-based tree method, which speeds up computation and helps regularization by controlling leaf splits (model complexity control).



Classification And Regression Trees (CART) is an algorithm of classification and regression tree analysis—a term used by (Leo & Friedman, 1984). In practice, it refers to the so-called “decision tree”. The tool presents various possible consequences (outcomes) of different scenarios and development paths. In the context of modeling, these scenarios are defined by dividing the data sample based on the values of the indicators within it. The levels and number of divisions are determined by minimizing the difference between the results of the “decision tree” and the real data. The importance of CART (Erden, 2023) is that it can measure the influence and importance of variables in the dataset during the construction of the tree structure. This can improve the accuracy of the model by identifying key variables. In addition, the method performs well with nonlinear datasets.

Using the tree structure to divide each individual dependent variable into subgroups (by defining a threshold value  $s$ ), the algorithm “searches” for the most homogeneous groups relative to the target value (Leo & Friedman, 1984):

$$R_1 = \{x \mid x_j \leq s\}, R_2 = \{x \mid x_j > s\} \quad (2)$$

The target function commonly used for classification tasks, such as the one in this study, is expressed using Gini impurity (Leo & Friedman, 1984):

$$G(R) = \sum_{k=1}^K p_k (1 - p_k) \quad (3)$$

This function measures how often a randomly selected element from the sample would be incorrectly classified based on the distribution of the data.

Classification and Regression Trees (CART) is a method, well known both in the literature and practice, and forms the foundation of some of the most modern and important ML algorithms, including Random Forest, XGBoost, Gradient Boosting, and others.

Gradient Boosting (GB), a gradient boosting method, belongs to the family of the so-called “ensemble learning” algorithms. This group of algorithms combines multiple weaker individual models in its architecture, often using the “decision tree” algorithm. In practice, the model builds on CART by creating multiple weak models and combining them to achieve better overall performance. This combination follows an iterative (additive) approach, where successive weak models are added to correct the errors of the previous iteration—a process known as “gradient boosting”. The method’s mathematical foundation is outlined below (Masui, 2022).

The target function measuring the difference between the real and predicted values:

$$\mathcal{L}(F) = \sum_{i=1}^n L(y_i, F(x_i)) \quad (4)$$

A new weak model is added at each subsequent iteration to improve the accuracy of the previous iteration ( $F_{m-1}(x)$ ). Optimization is achieved through “gradient descent” using the first derivative.

For  $m = 1$  to  $M$ :

$$r_{im} = - \left[ \frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right] \quad (5)$$

for  $I = 1, \dots, n$

$R_{jm} \exists j = 1, \dots, J_m$

$$\gamma_{jm} = \underset{\gamma}{\operatorname{argmin}} \sum L(y_i, F_{m-1}(x_i) + \gamma) \quad (6)$$

$$F_m(x) = F_{m-1}(x) + v \sum_{j=1}^m \gamma_{jm} 1(x \in R_{jm}) \quad (7)$$

Extreme Gradient Boosting (XGBoost) is one of the most powerful (Chen & Guestrin, 2016) and widely used ML algorithms today. In summary, the algorithm closely resembles Gradient Boosting and also belongs to the family of “ensemble learning” algorithms. XGBoost is optimized in terms of the speed of the hyperparameter optimization process, which is largely like that of Gradient Boosting. However, unlike Gradient Boosting, XGBoost includes built-in methods for regularization<sup>3</sup>.

The popularity of the algorithm is mainly due to its effectiveness and efficiency across a variety of real-world applications. It is suitable for both regression tasks and classification problems, such as fraud risk assessment (Dichev, 2023).

The algorithm is additive and builds “K” as a set of additive functions through decision trees (Chen & Guestrin, 2016):

$$\hat{y}_i = \phi(x_i) = \sum_{k=1}^K f_k(x_i) \quad (8)$$

The additive “training” of a model during iteration “t” has the following target function (Chen & Guestrin, 2016):

$$\mathcal{L}^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \quad (9)$$

The target function is further regulated by the expression for regularization ( $\Omega(f_t)$ ):

$$\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (10)$$

This component builds on Gradient Boosting. Through further optimization (Chen & Guestrin, 2016) of the target function, the authors introduce an optimized version of the function that utilizes both the first and the second derivatives. This, in turn, makes “model training” (i.e., hyperparameter optimization) a faster and more efficient process.

The brief descriptions of the methodology presented in this way highlight the advantages of applying fundamental and specialized analytical methods. This makes it possible to track their development and implement them as innovative statistical analyses and, at the same time, to efficiently build upon them to achieve the desired outcomes.

#### 4.2. Assessment of Discriminatory Power (Somers’ D/AUC)

In statistical analysis, this characteristic is used to measure the extent to which the model distinguishes between the two groups of transactions through its estimations. Scoring models are a good example because they aim to correctly rank individual clients based on their risk of default. An ideal model would distribute customers in such a way that it would be possible to separate all defaulted customers from non-default ones through the assessment of the model.

It is important to note that in the real world and practice, the observed data are often “imbalanced” (one of the two classes is more prevalent in the sample or population). In turn, this requires the use of appropriate metrics to accurately evaluate the effectiveness of the built fraud risk assessment tool (Dichev, 2023).

In this study, we address the challenge of evaluating the applied models on imbalanced datasets by using the Somers’ D metric. Unlike metrics such as accuracy, which can be highly misleading in the presence of class imbalance, Somers’ D provides a more reliable measure of a model’s discriminatory power. Somers’ D is an asymmetric measure of

the relationship between two variables that plays a central role as a rank parameter in “nonparametric” statistical methods. Given an independent variable  $X$  and a dependent variable  $Y$ , a score  $d_{Y/X}$  can be calculated as a measure of the effect of  $X$  on  $Y$ . That is,  $d_{Y/X}$  can be used as an indicator of how successfully  $X$  predicts  $Y$ . It is calculated by sorting the two variables ( $X$  and  $Y$ ) using the following expression (Göktaş & Öznur, 2011, p. 26):

$$d_{Y/X} = \frac{P - Q}{D_r} \quad (11)$$

$P = \sum_{i,j} f_{ij} C_{ij}$  = The probability that a randomly selected pair of observations will have an unambiguous order.

$Q = \sum_{i,j} f_{ij} D_{ij}$  = The probability that a randomly selected pair of observations will have opposite arrangements.

$C_{ij} = \sum_{k>i} \sum_{l>j} f_{kl} + \sum_{k<i} \sum_{l<j} f_{kl}$  = Total number of concordant pairs.

$D_{ij} = \sum_{k>i} \sum_{l<j} f_{kl} + \sum_{k<i} \sum_{l>j} f_{kl}$  = Total number of discordant pairs.

$f_{ij}$  = Total number of observations per row and column of table  $ij$

$$D_r = W^2 - \sum_{j=1}^C r_j^2 \quad (12)$$

$r_j$  = Total number of frequency allocations for row  $j$  of the double-sorted cross-table;

$W = \sum_{j=1}^C c_j = \sum_{i=1}^R r_i$  = Total observations in the sample.

The value of Somers'  $D$  is in the range from  $-1$  to  $1$ . A value of  $-1$  indicates a 100% negative relationship, while  $1$  signifies a 100% positive relationship between independent and dependent variables. A value of  $0$  indicates no relationship at all (Göktaş & Öznur, 2011, p. 26).

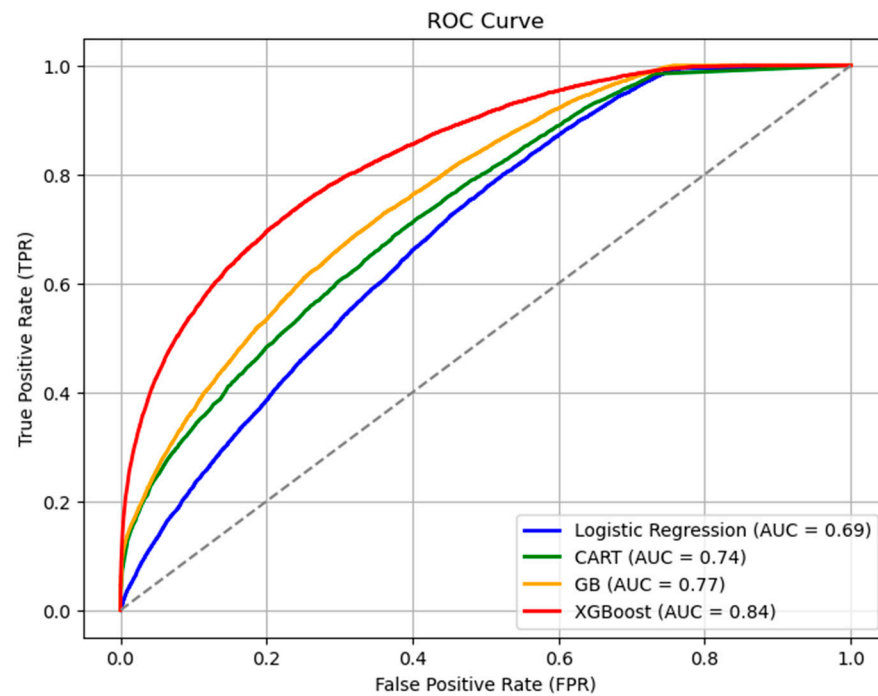
In this study, the Somers'  $D$  coefficient is used as the primary metric for discriminatory power. In the literature and practice, the metric commonly referred to as AUC (Area under the curve) is closely related to Somers'  $D$  (ECB, 2019, p. 68).

$$AUC = (d_{Y/X} + 1)/2 \quad (13)$$

For different types of models, the required and expected minimum level of discriminatory power varies. Given that in this study, the emphasis will be mainly on whether and to what extent ML methods contribute to a higher discriminatory power, we will consider the levels of the indicator compared to traditional algorithms such as logistic regression.

## 5. Analysis of Results

A variety of machine learning models, including Logistic Regression, (Classification and Regression Trees (CART), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost), were applied to explore the possibilities for managing the risk of fraud in bank transactions. As a result of the approbation performed, Figure 2 presents a comparative graph of the discriminatory power assessment of all applied algorithms (validation sample):



**Figure 2.** Comparative graph of the discriminatory power between all applied algorithms (validation sample). Source: Authors' calculations.

The results (Table 1) showed that in this sample, the applied ML algorithms outperformed logistic regression in terms of discriminatory power. The increase in discriminatory power is significant, especially for the best-performing algorithm (+0.29 Somers'D vs. logistic regression).

**Table 1.** Discriminatory power (test sample).

| Algorithm           | Somers'D (AUC) |
|---------------------|----------------|
| Logistic regression | 0.39 (0.69)    |
| CART                | 0.48 (0.74)    |
| GB                  | 0.54 (0.77)    |
| XGBoost             | 0.68 (0.84)    |

Source: Authors' calculations.

In the analysis of classification tasks, the errors of two types are considered: type I errors (false positives) and type II errors (false negatives). Type I errors refer to the number of observations (transactions) that are of a normal nature, but the model identifies them as fraudulent. Type II errors, on the other hand, represent the number of atypical transactions (frauds) that the model fails to detect and classifies as regular.

In risk management and business processes, different weights and significance can be assigned to the two types of errors, depending on the specific problem and the business model. For this analysis, equal weight has been assigned to both error types. The comparison between the two types of errors is presented in the following table.

Table 2 shows that XGBoost manages to improve both types of errors compared to logistic regression. Despite having a higher AUC for the rest of the applied ML methods, they improve significantly, but only in the first type of errors. These results are derived from the cutoff point that produces the best F1 score for each model individually, ensuring the most balanced trade-off between precision and recall for each. This highlights XGBoost's

advantage in balancing both error types, which may be critical in applications where both false positives and false negatives are costly.

**Table 2.** Errors of type I and type II (test sample).

| Algorithm           | False Positives | False Negatives |
|---------------------|-----------------|-----------------|
| Logistic regression | 23,415          | 6460            |
| CART                | 4072            | 6683            |
| GB                  | 3003            | 6694            |
| XGBoost             | 4367            | 5762            |

Source: Authors' calculations.

## 6. Implications for Research and Practice

While there are several studies on the use of machine learning (ML) for fraud detection and management, many of them lack a theoretical framework that helps track the evolution of algorithms or fail to highlight the key differences between them. Additionally, practical aspects are often overlooked, and in some cases, metrics like accuracy, which are not suitable for imbalanced datasets, are used to evaluate performance. This paper aims to address these gaps by providing a theoretical perspective on algorithm evolution, emphasizing practical aspects, and using more appropriate evaluation metrics for imbalanced data.

Based on the results, we can say that modern solutions in the field of artificial intelligence, particularly in machine learning, enhance classical approaches to risk assessment, expanding the ability to identify factors and manage emerging risks. These tools and their capacity to model large volumes of data are applicable in various areas, such as overall banking management, risk management, fast and efficient investment decision-making, front and back-office operations, and more. Artificial intelligence's capabilities allow for timely and accurate assessments, which can improve strategic decision-making. The role of artificial intelligence and advanced technologies is increasingly important in enhancing the security of banking transactions, and the banking sector is one of the fastest to adopt these value-added innovations.

From a business standpoint, the results presented in Table 2 indicate that XGBoost is more effective at detecting fraudulent transactions while minimizing the misclassification of legitimate transactions as fraud. As a result, this can lead to the following:

- Increased customer trust, as fewer of them will be affected by fraud transactions;
- An improved customer experience, with fewer disruptions in transactions;
- Reduced costs and losses for financial institutions due to more accurate fraud detection;
- Enhanced regulatory compliance by improving fraud prevention mechanisms;
- Greater operational efficiency by optimizing fraud detection processes;
- Others.

The tool can be integrated as a component of the fraud detection framework in financial institutions. Many financial institutions still rely on rule-based methodologies for fraud prevention in transactions. These rules are typically derived from past incidents of fraud and industry-wide information sharing. While this approach is widely recognized and applied, it has several limitations:

- The rule set tends to grow excessively large, making it difficult to understand, monitor, and maintain;
- Rules are often implemented after fraudulent activities have already occurred, meaning issues have already materialized;
- The approach is backward-looking, limiting its ability to identify new and unusual patterns of fraudulent behavior;

- Often, outdated rules persist due to the lack of regular evaluation of their adequacy and current applicability, resulting in a higher number of false positives.

A model could improve all of the aforementioned aspects. It can be implemented as an additional tool to proactively identify fraudulent transactions (ex-ante). Over time, it could even replace the rule-based approach, depending on how quickly and effectively a financial institution adopts and develops algorithmic solutions.

## 7. Conclusions

It is important to note that the results and conclusions drawn are based on the processing methods and data used in this study. Additionally, limitations arise from the transaction profile of specific financial institutions, as some institutions have very different profiles of transactor clients, meaning that the models and conclusions are limited to the data that have been used. Furthermore, the models may not generalize well to unseen data, as they are trained on a specific dataset that may not fully represent all possible transaction patterns. Additionally, new types of fraudulent transactions continuously emerge, making it crucial to monitor and update similar models regularly to ensure their effectiveness in business processes. Using different data and/or applying different processing techniques may lead to different results and conclusions. A critical factor is the already mentioned “feature engineering”, which plays a significant role in logistic regression, where the set of statistically significant factors is often limited. Identifying interactions and combinations between individual variables, creating new ones, and more can help improve the performance of all applied algorithms. The focus of the alternative methods is on the optimization of the hyperparameters. Their longer and more intensive optimization could further enhance results.

As a result of the study, the following key conclusions can be drawn:

First, the relevance of the topic is confirmed by the theoretical review of cutting-edge machine learning methods with high relevance for optimizing banking operations. In particular, it plays a significant role in risk assessment, detection of potential fraud, and its subsequent prevention, ensuring accuracy and efficiency in critical areas of the banking sector.

Second, the methodological analysis highlights the advantages of machine learning risk assessment methods that facilitate the processing of big data and offer innovative and highly productive solutions. This paper aims to compensate for the gaps in the methodological framework and neglected practical aspects by demonstrating an appropriate methodology based on unbalanced data.

Third, the validated models in this review show that the applied machine learning algorithms outperform less sophisticated models, such as logistic regression, in terms of discriminatory power. The improvement in discriminatory power is significant, especially for the best-performing algorithm (+0.29 Somers’ *D* vs. logistic regression).

Fourth, it is important to note that the reported improvement is based on a relatively large sample of observations and a limited set of algorithms. Using a different sample or incorporating additional algorithms could lead to different results. AI methods are more effective with large and unstructured datasets. Given the nature of the problem—bank transactions which occur in high frequency within financial institutions, a smaller sample to assess this effect would lack practical significance.

Fifth, based on the study, the relevance of the topic can be enriched through future research focused on methods of integration of hybrid models, bringing together additional different machine learning and deep learning models. The progressive methodological approach would include not only a combination of specialized artificial intelligence models but also a combined method of algorithms in hybrid structures, which would increase the



potential for more accurate predictions, overcoming the limitations of individual methods and increasing the relevance of their testing in practice.

The present study confirms the research hypothesis that applying an innovative and systematic approach to studying fraud in bank transactions through the development of a quantitative methodology based on modern and sophisticated machine learning algorithms positively contributes to the overall accuracy and effectiveness of risk management frameworks.

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## Abbreviations

The following abbreviations are used in this manuscript:

|         |                                     |
|---------|-------------------------------------|
| CART    | Classification and Regression Trees |
| GB      | Gradient Boosting                   |
| XGBoost | Extreme Gradient Boosting           |
| ML      | Machine learning                    |
| SVB     | Silicon Valley Bank                 |
| AI      | Artificial intelligence             |
| KNN     | K-nearest neighbors                 |
| VECM    | Vector Error Correction Model       |
| SVR     | Support vector regression           |
| WoE     | Weight of evidence                  |
| AUC     | Area under the curve                |

## Notes

- <sup>1</sup> More information: <https://www.jpmorgan.com/technology/artificial-intelligence/initiatives/synthetic-data/payments-data-for-fraud-detection> (accessed on 21 February 2025).
- <sup>2</sup> The synthetic database is provided by J.P. Morgan and is its exclusive property.
- <sup>3</sup> Methods for avoiding the so-called overfitting. See more about L1 (Lasso) and L2 (Ridge) regularization approaches.

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